

Starlink in the Wild: Multi-Perspective Measurements via DNS

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Abstract

Starlink, the low-Earth orbit (LEO) satellite constellation developed by SpaceX, has rapidly become the world's largest commercial satellite network and a key component of the global Internet infrastructure. Despite its growing prominence, critical aspects of its terrestrial operations, including its internal network architecture, user behavior patterns, and potentially vulnerable exposed services, remain largely unexamined. This paper presents a multi-perspective measurement that characterizes Starlink's ground-side infrastructure and ecosystem from a DNS-centric viewpoint. First, we leverage internal DNS leakage to infer the structure of Starlink's private networks. Second, we analyze passive DNS data to identify user behavior patterns and service usage trends. Finally, we perform large-scale active scanning of Starlink's IP address space to evaluate its service deployment and security posture, utilizing DNS records to isolate infrastructure-related IPs from those of end-users. Collectively, our findings provide novel insights into the architecture, operation, and security of Starlink's terrestrial network.

CCS Concepts

• **Networks** → **Network measurement; Network properties.**

Keywords

Satellite Internet; Starlink Ecosystem; Network Measurements; Network Structure; DNS

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1 Introduction

While advances in wireless access and fiber backhaul have expanded global connectivity over the past two decades [3, 11, 28, 44, 45], the emergence of low-Earth orbit (LEO) satellite networks marks a new phase in Internet access. Mega-constellations such as Starlink [55], OneWeb [19], and Project Kuiper [9] are deploying thousands of satellites at altitudes of 300–2000 km, enabling low-latency broadband coverage in regions underserved by terrestrial infrastructure. As a result, these systems are becoming global Internet Service Providers (ISPs) capable of bridging the digital divide [39, 58, 60], challenging terrestrial monopolies [56], and providing critical connectivity during disasters [27].

Among these initiatives, SpaceX's Starlink is the most ambitious and mature. As of early 2025, it operates the world's largest satellite constellation, with over 6,750 active satellites serving millions of users [59], and plans to expand to approximately 42,000 satellites [24]. This scale has drawn substantial academic and industry attention; however, prior work has largely focused on the space segment, including performance evaluation, inter-satellite topology modeling, and routing design [43, 67]. As a result, Starlink's terrestrial components remain poorly understood, limiting a comprehensive end-to-end view of its network architecture and operations.

Current research overlooks several vital aspects of Starlink's ground-side infrastructure and user ecosystem. *First*, while prior work has analyzed Starlink's backbone topology [46, 64], the internal network structure of its Points of Presence (PoPs) has not been systematically studied. *Second*, there is a lack of research using passive data sources to analyze real-world Starlink user access

patterns, leaving a void in our understanding of usage trends and regional traffic dynamics. *Finally*, the nature of its exposed network assets—including open ports, running services, and application versions on both infrastructure and user devices—remains unclear, impeding thorough security analysis and risk assessment.

To address these gaps, this paper presents a multi-perspective measurement study of Starlink’s terrestrial infrastructure and user ecosystem using a methodology that combines DNS leakage analysis, passive DNS (PDNS) record analysis, and large-scale network asset fingerprinting. These methods enable us to reconstruct hidden internal structures, analyze user activity patterns, and evaluate the exposure of publicly reachable assets.

Network Structure Inference. First, we reconstruct Starlink’s internal network architecture by detecting and analyzing private DNS record leakages. This approach allows us to map the internal network hierarchies within Starlink PoPs, revealing consistent structural patterns across its global infrastructure (Section 4).

Characterizing Starlink’s User Behavior. Second, we characterize Starlink’s global user behavior by analyzing billions of queries from a large-scale PDNS dataset. This analysis uncovers rich geographic and temporal access patterns, identifies popular services, and reveals event-driven traffic dynamics. (Section 5).

Asset Measurement and Vulnerability Analysis. Finally, we perform a comprehensive security assessment by scanning and fingerprinting publicly exposed Starlink network assets, and classify them into user assets and infrastructure assets based on DNS records. We identify over one million asset entries and, by cross-referencing them with vulnerability databases, uncover widespread security risks, including legacy software deployments, known CVEs, and misconfigured services (Section 6).

In summary, this study provides a novel and systematic analysis of Starlink’s terrestrial architecture, user base, and security posture. Our key contributions include: (i) the first systematic reconstruction of Starlink’s internal network architecture, (ii) a detailed understanding of user behavior at a global scale, and (iii) a security-focused assessment of Starlink’s exposed network assets. These findings offer a foundation for future work on satellite network measurement, optimization, and cybersecurity. A subset of our privacy-preserving dataset is available at <https://doi.org/10.5281/zenodo.17221250>.

2 Measurement Objectives and Methodology

This section presents our investigation framework. We introduce Starlink as the measurement target, then outline the study objectives and the methodologies used to achieve them.

2.1 Starlink

Starlink provides low latency, high bandwidth, wide coverage, and flexible deployment, making it well suited for bridging the digital divide and enhancing connectivity resilience. Most satellites operate in a 53° orbital shell serving mid-latitudes, with additional 70° and 97.6° shells extending polar coverage [43]. Its architecture includes five components: the satellite constellation, user terminals, ground stations (GSs), PoPs, and inter-satellite laser links (ISLs). Users connect via the “Dishy” Ku-band phased-array terminal, while satellites

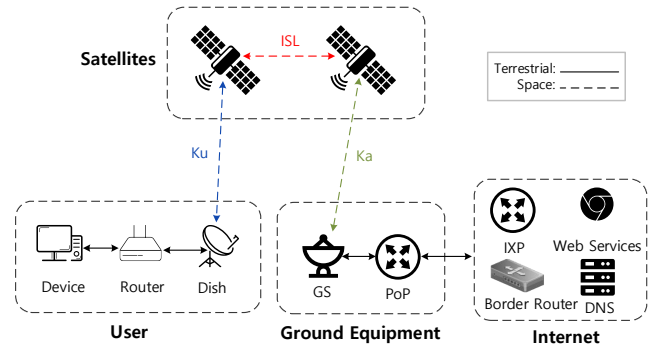


Figure 1: An overview of the Starlink network architecture.

employ multiple antennas to form subdivided beams serving many terminals simultaneously.

Traffic follows two routing modes: direct *bent-pipe* links when a terminal and GS share satellite coverage, or *extended bent-pipe* paths using ISLs across satellites when they do not [43]. Traffic then reaches a GS, is forwarded to a Starlink PoP, and exits to the Internet. Our study focuses on Starlink’s terrestrial components: PoPs, user behavior, and network assets—and uses the official *Starlink GeoIP Feed* for geographic attribution [30, 57].

2.2 Measurement Objectives

This study pursues three primary measurement objectives designed to address the key gaps in the existing literature on Starlink’s terrestrial network.

Network Structure Inference. Given limited public documentation, we seek to reconstruct Starlink’s ground infrastructure by identifying exposed components (e.g., routers and service endpoints) through DNS leakage and applying semantic inference to model the internal topology.

User Behavior Analysis. To better understand Starlink user activity, we examine network interaction patterns at both macro and micro levels, including geographic distribution, temporal dynamics, and service preferences. Time-series and event-driven analyses are used to detect anomalies related to regional events or disruptions.

Asset Discovery and Security Assessment. To evaluate the security posture of the Starlink network, we perform asset discovery to identify publicly accessible assets, which we classify into user and infrastructure assets. By analyzing open ports, service banners, and device fingerprints and correlating them with vulnerability databases, we identify potential security risks.

2.3 Measurement Methodology

Our study uses a multifaceted methodology that combines both passive and active measurements. Since our entire approach is DNS-centric, we began with a preliminary investigation of Starlink’s DNS performance to provide a reference for our DNS selection. A summary of this initial analysis is available in Appendix F.

2.3.1 Target Collection. Our measurement targets include IPv4 and IPv6 address spaces associated with Starlink’s Autonomous

Systems (ASes), AS14593, AS27277 and AS45700. For IPv4, we aggregated prefixes from authoritative BGP sources [4, 23, 25] and integrated data from official GeoIP Feed [57] to ensure comprehensive coverage. For IPv6, where exhaustive scanning is impractical, we adopted a targeted discovery approach based on the methodology of Pan et al. [64], supplemented with active addresses from public IPv6 hitlists [18] to create a representative measurement set.

2.3.2 Active Measurement. We employed several active measurement techniques to probe the Starlink network.

IP Liveness Detection and Feature Extraction. We used XMap [36] for large-scale probing to identify active hosts in our target address sets. For each responsive IP, we performed reverse DNS lookups to retrieve PTR records. These records were then semantically analyzed to infer functional roles and locations based on naming conventions (e.g., `customer.bgtacol1.pop.starlinkisp.net` indicating a PoP in Bogotá, Colombia). By generalizing across such patterns, we extracted naming conventions that facilitated subsequent structure inference.

Network Structure Inference. Our approach to inferring internal network structure combines passive discovery with active probing. We first identified DNS leakage in PDNS records, where publicly resolvable hostnames map to private RFC1918 [50] address space. Using the naming patterns observed in these leaked records, we algorithmically generated additional candidate hostnames and issued active DNS queries and scans to uncover more private hostnames and IP addresses. Finally, we applied heuristic semantic analysis to the collected data to reconstruct the internal network structure.

Asset Measurement and Vulnerability Analysis. We fingerprinted active hosts to create an inventory of Starlink’s public-facing services. Our scans detected open TCP/UDP ports, identified running services and protocols, and extracted OS-level signatures. The scans prioritized ports associated with sensitive or administrative services, including 22 (SSH), 23 (Telnet), 20/21 (FTP), 443 (HTTPS), 7547 (TR-069), and 161 (SNMP). Identified services were then cross-referenced against public vulnerability databases (e.g., NVD, CVE) to assess their security posture.

2.3.3 Passive Measurement. To capture real-world user activity, we analyzed PDNS logs collected from public resolvers. This dataset, encompassing billions of DNS resolution events over an extended period, enabled us to characterize long-term user access trends and service interactions. We built a distributed processing pipeline using HDFS and Apache Spark, allowing scalable ingestion, filtering, and multidimensional analysis. The results reveal both macro-level user activity trends and micro-level service access patterns.

3 Dataset Overview

This section outlines our data collection and processing methods and summarizes the four datasets used in this study. All datasets are derived from or enriched with DNS-related data: a *foundational* dataset for PoP analysis, a *structure inference* dataset from DNS leakage, a *PDNS* dataset for user behavior analysis, and an *asset scanning* dataset for security assessment.

Foundational Dataset (Nov 2024 – Mar 2025). We conducted a five-month longitudinal PTR scan of over 300,000 IPv4 addresses to examine the evolution and geographic distribution of Starlink PoPs.

Extracted semantic features (e.g., PoP identifiers) support target identification and naming pattern analysis. We identified 60 unique fingerprints, with multiple ones corresponding to the same PoP. These fingerprints are mainly concentrated in North America, with fewer in Africa. Recent deployments such as `jnhgzaf1` nevertheless highlight continued expansion into underserved regions [61].

Structure Inference Dataset (Dec 2024 – Apr 2025). This dataset comprises leaked internal DNS records aggregated by IP prefix and annotated with semantic tags to infer Starlink’s internal network structure. It includes 3,018 records, of which 2,668 resolve to private IPs and 350 to public IPs (see Appendix C).

PDNS Dataset (Nov 2023 – Mar 2025; Jul 2025). We collected 4.3 billion passive DNS logs (≈ 260 GB), grouped by IP, date, and second-level domain, and enriched them with Starlink GeoIP data [57] and PoP fingerprints. Comparisons between early (Nov 2023–Jan 2024) and later periods (Jan–Mar 2025) reveal notable regional shifts in user activity (see Figure 8 in Appendix B).

Asset Scanning Dataset (Feb 2024 – Mar 2025). This dataset contains over 1.5 million deduplicated records from active scans, capturing publicly exposed Starlink infrastructure and user devices. Records were normalized, stored in JSON format, and tagged with timestamps, PoP fingerprints, and geolocation metadata. Analysis indicates a large presence of IoT devices, particularly video surveillance systems (see Figure 9 in Appendix E).

Privacy Processing. All potentially identifying IP addresses were anonymized using the CryptoPan algorithm [13] (see Appendix A).

4 Network Structure Inference

This section presents a heuristic methodology for reconstructing the internal architecture of the Starlink network. We first identify and validate internal DNS leakage to discover otherwise hidden domain names (Section 4.1) and then leverage this leakage to infer the structure of internal network components (Section 4.2).

4.1 Identifying and Validating DNS Leakage

Analysis of our PDNS dataset revealed that certain Starlink hostnames, which are systematically named but resolve to private IP addresses, unexpectedly appear in public recursive DNS traffic. To confirm this “leakage,” we issued direct queries against Starlink’s authoritative server (`dns31.cloudns.net`). Although these hostnames should logically resolve only within Starlink’s private namespace, they were publicly accessible, suggesting misconfigurations or deliberate exposure (e.g., via VPN, NAT64, or zone-file overrides). Such leakage blurs the boundary between internal and external DNS, posing potential security risks. We disclosed our findings to SpaceX’s security team, *which confirmed the issue and indicated that the affected records may be migrated to non-public DNS infrastructure as a future mitigation*.

Semantic Extraction of Naming Patterns. Building on these observations, we harvested all analogous domain names from the PDNS dataset and clustered them by their PoP fingerprints (identified in foundational dataset in Section 3). Through string analysis, we extracted naming patterns, summarized in Table 1, identifying both intuitive labels (e.g., `mgmt`, `agg`, `edge`) and more obscure tags (`spi`, `spx`, `pdu`, `pkp`). To interpret the latter, we applied a three-step heuristic combining *contextual device naming*, *standard*

Table 1: Frequency of keywords in Starlink hostnames.

Naming scheme	Inferred meaning	Percentage
agg	Aggregation	6.33%
edg	Edge	3.71%
mgt	Management	37.48%
oob	Out of Band	6.63%
pkp	Packet Processing	31.28%
spi	Service Plane Internal	5.63%
spx	Service Plane External	3.55%
con	Controller	4.47%
fwl	Firewall	6.86%
nbz	Network Boundary Zone	6.23%
pdu	Power Distribution Unit	15.67%
rtr	Router	18.42%
srv	Server	37.44%
swc	Switch	6.39%
tor	Top of Rack Switch	2.39%

networking terminology, and *logical consistency*. A detailed walk-through of this process is provided in Appendix D. We also noted emergent patterns such as *iot*, *vpn*, and *gws*, which may indicate forthcoming service expansions.

Domain Generation and Resolution Campaign. Leveraging the extracted patterns, we algorithmically generated approximately 4 million candidate hostnames. Between December 2024 and April 2025, we performed weekly DNS resolution scans, throttled at 100 queries per second to minimize the impact on DNS infrastructure. This campaign yielded 3,018 unique *domain-to-IP* mappings: 2,668 (88.3%) mapped to private addresses, while 350 (11.7%) resolved to public IPs. Notably, 34.1% of these public addresses fell outside of Starlink’s primary Autonomous Systems (AS14593 and AS45700), suggesting that a significant fraction of its services is hosted on third-party infrastructure.

Functional Insights from Naming Patterns. Table 1 reports the distribution of discovered naming patterns. The dominance of *mgt* and *pkp* labels indicates extensive deployment of management and packet-processing services, supporting automated remote operations and load-balancing functions. Significant counts of *srv*, *rtr*, and *pdu* labels reveal a robust physical infrastructure, including numerous servers for data processing, routers for traffic management, and power distribution units for operational reliability. These findings provide the first empirical glimpse into Starlink’s private network components, laying the groundwork for our subsequent structure reconstruction.

4.2 Inferring Internal Network Structure

Building upon our DNS-leakage analysis, we developed a heuristic methodology to reconstruct the internal structure of a Starlink PoP. We found that all PoPs appear to share a similar, modular design; accordingly, we use the largest European PoP, *frntdeu1*, as a representative case study.

Within *frntdeu1*, devices are organized into five functionally identical “units,” each numbered 1 through 5. Every unit comprises the following components:

- **Out-of-Band Management Components:** an out-of-band controller (*oob-con*) and two firewalls (*mgt-fw1-oob*, *mgt-fw1*) to ensure remote access even if the primary network is unavailable;
- **Management Infrastructure:** management switches (*mgt-swc*), routers (*mgt-rtr*), and redundant power distribution units (*mgt-pdu a/b*) for centralized control and power resiliency;
- **Core Routing and Aggregation:** aggregation routers (*agg-rtr*) that collect traffic from edge and service-plane routers;
- **Service Processing:** management servers (*mgt-srv*) and packet-processing servers (*pkp-srv*) responsible for tasks such as load balancing, session management, and packet forwarding.

This modular, scale-out architecture, in which each unit is self-contained, facilitates partitioned management, traffic isolation, and localized fault handling, embodying the high-availability and scalability principles of carrier-grade PoPs.

By correlating naming conventions with IP allocations, we logically partitioned *frntdeu1* into four zones, shown in Figure 2:

- (1) **Out-of-Band Management Zone** (e.g., *mgt-fw1-oob*): Provides an independent management path separate from the service network. Dual firewalls enhance security for both this zone and the adjacent Management Zone;
- (2) **Management Zone** (172.17.12.*): Hosts *mgt-swc*, *mgt-rtr*, and *mgt-pdu* devices. We identify a subcomponent, *mgt-nbz* (“network boundary zone”), which, alongside *pdus*, likely enforces physical isolation and fault detection on critical links;
- (3) **Service Forwarding Zone** (172.17.12.*): Contains *edg-rtr*, *spi-rtr* (service-plane internal router), and *agg-rtr*, which collectively aggregate and steer user traffic toward the processing cluster;
- (4) **Service Processing Zone** (172.19.4.*): Isolated from management traffic, this zone hosts the *mgt-srv* and eight *pkp-srv* nodes, forming a scalable processing cluster for tasks such as caching, packet handling, and load balancing.

Our methodology is inspired by prior work that leveraged DNS naming patterns for functional and structural network inference [2, 68]. Although public traceroutes cannot validate paths to private IP addresses, our DNS-based analysis provides the first systematic model of Starlink’s PoP internals. To strengthen our findings, we submitted the inferred mappings to a large language model for independent interpretation, which reached conclusions consistent with our own.

Takeaway #1 - By leveraging DNS leakage and semantic naming patterns, we reconstruct Starlink’s internal network structure, revealing that PoPs follow a standardized, modular architecture. A case study of *frntdeu1* highlights distinct management, forwarding, and processing zones, offering new insights into the operational design of Starlink’s terrestrial infrastructure.

5 Characterizing Starlink’s User Behavior

This section presents a multidimensional analysis of PDNS data to characterize Starlink user behavior, focusing on service preferences, query volume trends, resolution failure patterns, and global outages. While our measurement vantage points may introduce geographic bias, the dataset still provides valuable insights into representative real-world activity across the Starlink network.

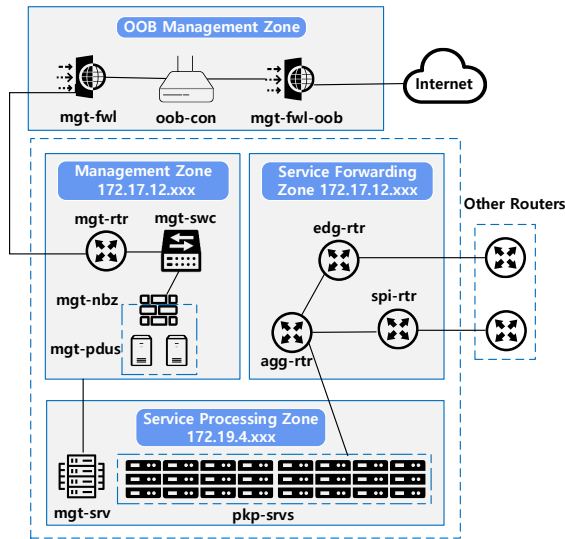


Figure 2: The inferred internal network structure of the Frankfurt PoP.

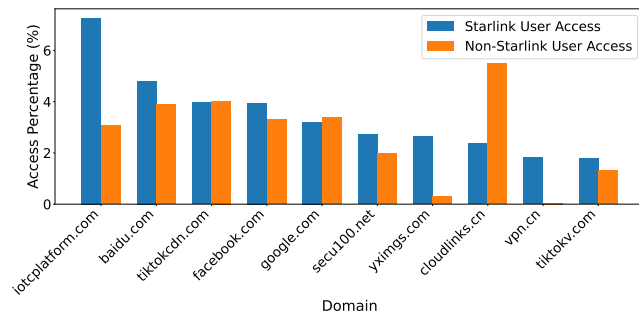


Figure 3: A comparison of service access preferences between Starlink and non-Starlink users.

5.1 Accessed Services and Domain Patterns

To characterize service usage, we analyzed the second-level domains (SLDs) accessed by Starlink users. We used SLDs rather than fully qualified domain names (FQDNs) to preserve user privacy and reduce data processing complexity. To highlight distinctive traffic patterns, we compared Starlink’s domain access profile against that of non-Starlink users from the same PDNS dataset.

One notable distinction is the high prevalence of queries to `iotcplatform.com` among Starlink users, which accounts for approximately 7.3% of total queries to the top 1,000 SLDs, a significantly higher proportion than observed among non-Starlink users, as shown in Figure 3. *This finding indicates widespread adoption of Internet of Things (IoT) services on the Starlink network and is corroborated by our active scanning results (see Section 6.1 for details).*

We further analyzed regional service preferences by ranking the top 15 SLDs accessed by Starlink users in select countries on every continent except Antarctica. As shown in Figure 4, the results reveal significant geographic variation in service adoption. For example, cloud service domains such as `myhuaweicloud.com` and

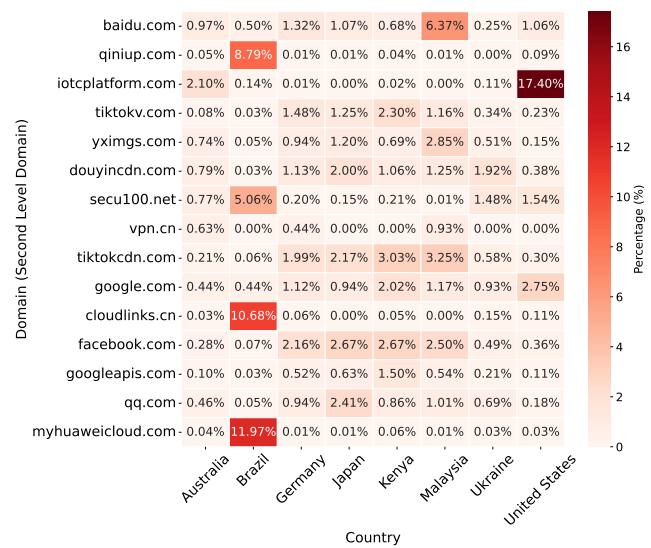


Figure 4: Geographic variation in service access among Starlink users.

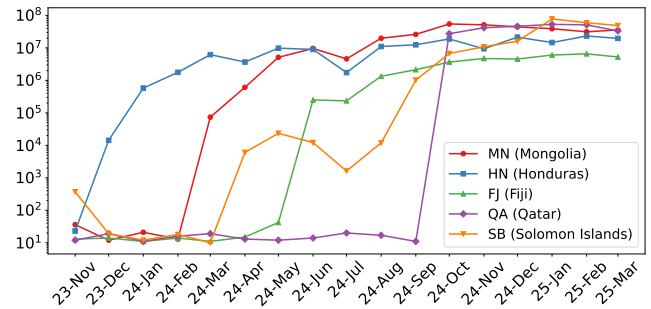


Figure 5: Temporal trends in Starlink DNS query volume across selected countries.

`cloudlinks.cn` are highly prevalent in *Brazil* but nearly absent elsewhere. Similarly, `iotcplatform.com` shows substantial activity in the United States but is negligible in other regions.

5.2 Temporal and Geographic Access Trends

By analyzing labeled PDNS data over time, we can infer the temporal and geographic rollout of Starlink services. In regions such as *Mongolia*, *Honduras*, *Fiji*, and *Qatar*, we observed sharp increases in DNS query volumes from near-zero baselines, as shown in Figure 5. These trends closely coincide with publicly documented service launch dates, as corroborated by records from sources like Wikipedia [65]. We did, however, identify some anomalies; for instance, the *Solomon Islands* exhibited transient Starlink traffic before its official launch in September 2024. Such discrepancies are likely attributable to inaccuracies in GeoIP Feed [57], which can misclassify IP geolocations for satellite-based connections.

To assess usage intensity, we ranked countries by total query volume, identifying *Malaysia*, the *United States*, the *Philippines*,

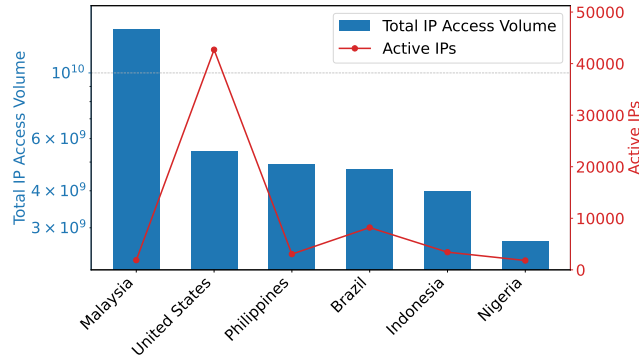


Figure 6: Assessing Starlink usage intensity in the highest-traffic countries.

Brazil, and Nigeria as the top five contributors (Figure 6). Many of these countries have limited terrestrial Internet infrastructure or significant maritime regions, making satellite connectivity an attractive alternative. In particular, in Malaysia, the Philippines, and Nigeria, we observed high query volumes originating from a relatively small number of unique source IPs. This pattern strongly suggests the prevalence of large-scale NAT, indicating that Starlink connections are frequently shared across households or communities.

5.3 Resolution Failures

Our analysis of DNS queries directed at Starlink-related domains (e.g., *.starlinkisp.net, *.starlink.com) revealed a resolution failure rate exceeding 70%. We identified two primary causes for these failures:

Private-only Domains. As discussed in Section 4.1, a significant portion of failed queries targets internal domains that should not be exposed to the public DNS. For example, requests for dishy.starlink.com and domains under starlinkisp.net accounted for 61.1% and 37.7% of these private queries, respectively. These domains are intended for internal resolution but appear in public DNS traffic due to factors such as client misconfiguration, VPN usage, NAT64 interactions, or zonefile overrides [47], creating operational complexity and information exposure risks.

Misused Hostnames. We also observed numerous queries for hostnames that incorrectly treat Starlink PoP fingerprints as resolvable domains (e.g., attempting to resolve jtnaidn2). Hostnames such as customer.sfiabgr1.pop.starlinkisp.net are merely identifiers and are not meant to resolve to a routable IP address. Queries for such malformed hostnames had a resolution success rate of only 0.3%, indicating widespread misbehavior by clients or scripts in parsing PoP identifiers.

5.4 Analysis of a Global Starlink Outage

Our analysis of passive DNS (PDNS) data captured a global Starlink outage that occurred on July 24, 2025, between 19:16 and 21:28 UTC. During this event, DNS query traffic across all PoPs dropped sharply and simultaneously, followed by a gradual recovery. Figure 7 illustrates this synchronized disruption in network activity across several major PoPs.

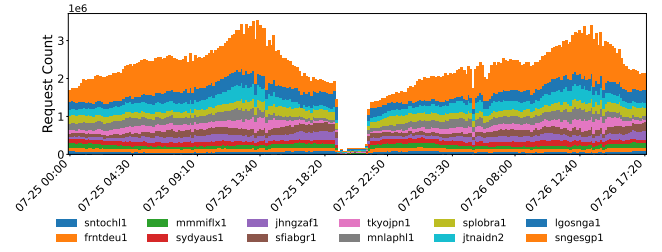


Figure 7: DNS query volume over time for major Starlink PoPs.

Lasting over two hours, the incident represents the most severe outage in Starlink’s history. Public reports suggested it was triggered by a critical internal software failure, resulting in a worldwide service disruption [33]. The outage had widespread consequences, affecting communications in Ukraine, mining and maritime operations, the Starshield defense service, and industrial IoT connectivity [20]. This event highlights the systemic risks and vulnerabilities inherent in global-scale satellite internet infrastructure.

Takeaway #2 — Passive DNS analysis uncovers distinctive temporal and regional patterns in Starlink usage. We find widespread IoT-related activity, heavy NAT-induced query aggregation, and strong regional biases in service access. The emergence of traffic in new regions closely tracks official service rollouts, whereas anomalies, such as traffic from areas without announced coverage, expose inaccuracies in the GeoIP Feed [57] and highlight the need for its continuous maintenance by Starlink. Misconfigured clients frequently leak internal-only domains, and more than 70% of Starlink-related queries result in resolution failures. Our monitoring also captures a major global outage, during which all PoPs experienced a synchronized traffic collapse lasting over two hours. Together, these observations highlight the operational complexity and heterogeneous dynamics of the Starlink ecosystem.

6 Asset Measurement and Analysis

This section presents the findings from our large-scale measurements of Starlink’s network assets. We scanned over 300,000 IPv4 addresses across more than 3,000 ports to identify assets exposed via common protocols such as FTP, SNMP, SMTP, Telnet, and HTTP/HTTPS. Using automated scanning tools in conjunction with a pre-compiled knowledge base, we extracted information from service banners, HTTP response bodies, and headers to infer device types, protocols, applications, and version metadata.

6.1 Asset Discovery

After deduplication and aggregation, our active scans identified approximately 64,000 responsive IP addresses, spanning 1,942 ports, 154 protocols, and 414 unique service types. This corresponded to a liveness rate of 20.4% for Starlink’s IPv4 address space. In contrast, our passive DNS records revealed around 140,000 IPs with more than 10 queries, suggesting a much higher liveness rate of 44.4%. This discrepancy indicates that many devices are observable via passive signals but do not respond to direct scanning. By combining both active and passive datasets, we constructed a comprehensive

Table 2: The top 10 most frequently observed ports and protocols among Starlink network assets.

Port	Proportion	Protocol	Proportion
500	12.92%	HTTP	19.00%
443	8.13%	HTTPS	15.19%
161	6.51%	SIP	8.53%
80	6.04%	SSH	7.43%
22	4.05%	ISAKMP	6.75%
5060	2.97%	IKEv2	6.12%
5061	2.91%	SNMPv3	4.56%
1701	2.89%	L2TPv2	2.88%
6060	2.83%	MGCP	2.81%
2427	2.82%	BW Test	2.76%
Others	47.93%	Others	23.97%

Table 3: Exposure of sensitive network protocols across scanned Starlink IP addresses.

Protocol	Telnet	POP3	SNMPv3	NetBIOS	FTP	SMTP	SNMP	HTTP
IP Share	2.83%	0.13%	15.41%	0.38%	3.13%	1.16%	3.58%	34.18%

inventory of approximately 180,000 unique live IP addresses. This measurement scope is significantly broader than that of prior work, such as Seestar [66], which reported 10,000 live IPs.

The top 10 most frequently observed protocols and ports are summarized in Table 2. While web services like HTTP and HTTPS dominate the protocol landscape, their default ports (80 and 443) are less prominent in the port rankings, implying that many services operate on non-standard ports. We also observed a notable number of sensitive protocols, including Telnet, FTP, SMTP, and SNMP, which are known to pose security risks due to plaintext communication and potential credential leakage. Table 3 summarizes the proportion of IPs exposing these vulnerable services.

Our analysis of asset distribution by PoP provides an approximate view of each location’s operational footprint. The *Jakarta* PoP (jtnaidn2), for example, showed a particularly high number of assets. As noted in other studies [64], *Jakarta* has limited IPv6 adoption, leading to heavier reliance on IPv4 and thus higher asset visibility. Conversely, a city like *Seattle* (sttlwax1), which has a large user base and broader IPv6 adoption, exhibited fewer IPv4-exposed assets. This is corroborated by our PDNS data, where the *Jakarta* PoP logged over 3.9 billion queries compared to *Seattle*’s 500 million, highlighting the relative intensity of IPv4 activity. The discovered assets span a broad range of categories (e.g., operating systems, IoT devices, security appliances, and VPN products), with IoT devices constituting a disproportionately large share, consistent with the behavioral patterns observed in our PDNS analysis. A detailed breakdown of asset types is provided in Table 4.

6.2 Asset Classification via DNS Records

As discussed in Section 4 and prior work [68], DNS records often contain rich semantic cues that can reveal the functional roles of network assets. Therefore, we combined *DNS semantics*, *WHOIS*

Table 4: Distribution of discovered Starlink assets.

Asset Type	Proportion
Development Frameworks	17.59%
Scripting Languages	16.98%
Operating Systems	13.70%
ADSL	12.31%
Routers	9.47%
Security Products	6.90%
Video Surveillance	3.41%
Traffic Analysis	2.81%
VPN Products	2.35%
Office Systems	2.29%
Web Servers	2.16%
Others	10.30%

data, and *asset scan results* to distinguish between user assets and infrastructure assets.

1. Establish a Seed List from DNS Records. First, we established a seed list by collecting 8,461 Starlink domains from PDNS logs, certificate transparency records [53], and our domain generation campaign (Section 4). These domains resolved to 1,605 unique public IP addresses, as multiple domains can map to the same IP.

2. Expanding the List with Whois Records. Next, we expanded this list using Whois data, leveraging the common ISP practice of separating user and infrastructure address pools [22]. We queried our seed IPs and used the NetRange and NetName fields (e.g., SPACEX-STARLINK-IPV4) to identify their parent CIDR blocks. Based on the heuristic that a single infrastructure IP classifies its entire parent block, we identified 79 infrastructure subnets. These included critical ranges like 206.224.64.0/19, hosting backbone routers [46] and firewalls (e.g., sttlwax1-sat-fw11.starlinkisp.net).

3. Cross-Validate with Asset Scanning. Finally, we cross-validated these classifications by analyzing our asset scan data. We confirmed that the identified IP blocks predominantly host services typical of network infrastructure, such as SIP, MGCP, and SSH endpoints, while some host CDN-related services (e.g., 206.214.231.0/24) for providers like Google and Facebook, further supporting their classification as infrastructure.

6.3 Vulnerability Identification

We performed fuzzy vulnerability matching using device metadata and service banners. By correlating Common Platform Enumeration (CPE) identifiers with public vulnerability databases, we identified 12,675 vulnerable assets across 7,519 unique IPs, associated with 45,906 vulnerabilities spanning 550 unique CVEs. This indicates that 16.4% of all live IPs captured in our active scans have at least one known vulnerability.

The severity distribution, summarized in Table 5, reveals that High and Critical vulnerabilities together account for over 50% of all scored findings. A relatively large proportion of entries are labeled as None, primarily because certain vulnerabilities have not yet been assigned official NVD scores. The prevalence of vulnerabilities associated with outdated components, such as dnsmasq, MikroTik RouterOS, OpenSSH, and PHP, indicates that a substantial

Table 5: Distribution of vulnerabilities by severity levels.

CVSS v3 Score	Severity	Count (%)	Example CVE
0	None	11812 (25.73%)	CVE-2014-0109
0.1–3.9	Low	111 (0.24%)	CVE-2024-10452
4.0–6.9	Medium	10810 (23.55%)	CVE-2021-32786
7.0–8.9	High	14758 (32.15%)	CVE-2023-31122
9.0–10.0	Critical	8415 (18.33%)	CVE-2022-23943

number of Starlink-exposed assets are affected by unpatched or misconfigured software, posing significant security risks.

In addition, we observed widespread TLS certificate issues. Of the 23,512 certificates we collected, over 90% were either self-signed or issued by private certificate authorities. While this practice is common for IoT devices and network appliances, these certificates are not trusted on the public Internet. We also detected the use of insecure cipher suites, such as those relying on 3DES or RC4, which expose services to known cryptographic vulnerabilities.

Our IPv6 asset measurement adopted the approach of Pan et al. [64] and utilized hitlist data [18] to build a Starlink-specific target list. Active scanning of this set revealed only 3,478 assets on 1,344 unique IPs across 113 ports, the majority of which were web-related services.

Takeaway #3 - Our large-scale asset scanning reveals that Starlink exposes a diverse set of network devices, many of which rely on sensitive protocols or operate with outdated software. Alarming, 16.4% of active IPv4 addresses discovered through scanning are associated with known vulnerabilities, and over 90% of observed TLS certificates are self-signed or privately issued. These results underscore significant security risks stemming from misconfigurations, insufficient patching, and the lack of standardized security enforcement across the Starlink network.

7 Limitations and Future Work

While this study provides a comprehensive external view of Starlink’s terrestrial network, several limitations remain and motivate future work.

Internal Network Structure Inference. Our reconstruction of Starlink’s internal network is based on roughly 3,000 leaked domain–IP mappings and therefore offers only a partial view. Future work will incorporate richer features and more advanced inference techniques to derive a more complete architectural model.

Asset Measurement and Analysis. (a) Our asset discovery relies on active scanning and cannot capture non-responsive devices. Future efforts will integrate PDNS-derived behavioral features to improve large-scale asset identification, particularly for IoT devices [38, 49]. (b) Although key Starlink infrastructure ranges (e.g., 206.224.64.0/19) responded to probes, service-level fingerprints remained limited. Future work will explore more advanced probing techniques and expanded fingerprint databases to improve identification [1]. (c) IPv6 analysis is constrained by the vast address space, making exhaustive scanning impractical. Future research will generate candidate IPv6 targets and leverage broader hitlists to improve coverage [52].

8 Related Work

A growing body of work has examined the Starlink network from multiple perspectives. Several studies evaluate performance, measuring latency and throughput from global vantage points [16, 37, 42], analyzing application-level behavior [29, 69], and comparing Starlink with terrestrial cellular networks [21, 39]. Other efforts rely on exposed services or crowdsourced data to characterize performance and service quality [26, 43].

Beyond performance, fewer studies investigate Starlink’s internal or backbone architecture. Tanveer et al. [62] highlighted the role of a global controller in satellite reconfiguration, while traceroute-based analyses inferred backbone topology and routing behavior [46, 64]. Simulation-based work further explores routing strategies and network optimization in LEO constellations [5, 6, 32].

Meanwhile, passive traffic studies have focused on terrestrial wired networks [7, 17, 40, 63] or backbone traffic across heterogeneous access types [8, 31]. While PDNS data has been used for device identification and anomaly detection [12, 38, 49], satellite-specific passive analyses remain limited. A notable exception is the work of Trevisan et al. [63], who examined user behavior using monitors deployed at GEO satellite ground stations.

In the area of asset discovery, prior work has employed techniques such as active scanning [14], machine learning–based classification [41], and fingerprinting optimization [35]. Platforms like FOFA [15] and Shodan [54] aggregate large volumes of asset data, but only SeeStar [66] provides a dedicated, systematic measurement platform focused on Starlink.

In summary, while academic interest in Starlink is growing, existing studies have primarily concentrated on performance evaluation, topology inference, or routing analysis. In contrast, our work offers the first comprehensive, multi-perspective study of Starlink’s terrestrial infrastructure, encompassing its internal network structure, global user behavior, and exposed asset landscape.

9 Conclusion

This study presents the first multi-perspective measurement of Starlink’s terrestrial network architecture and ecosystem via DNS from external vantage points. By combining active and passive techniques, we analyze Starlink’s internal structure, user activity, and exposed assets. Our findings uncover significant internal information leakage via DNS, identify trends corresponding to global service events, and document widespread IoT deployments with substantial security risks. These results provide novel visibility into Starlink’s operational footprint and its integration with the global Internet. More broadly, our work demonstrates the value of DNS-based analysis for characterizing large-scale satellite networks and establishes a foundation for future research on their measurement, optimization, and security.

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A Ethics

All measurements in this study follow responsible Internet measurement practices and are designed to minimize potential risks to users and network infrastructure.

Active measurements are carefully rate- and frequency-limited to reduce operational impact. PTR record scans targeting Starlink IP addresses are capped at 100 queries per second and conducted no more than three times per week, while DNS leakage scans follow the same rate limit and are performed at most once per week. Upon discovering the leakage issue, we responsibly disclosed our findings through SpaceX's official Bugcrowd vulnerability disclosure program [10] and received confirmation from their security team. To the best of our knowledge, the issue remains unresolved at the time of writing.

Asset measurements are restricted to non-intrusive techniques, such as banner grabbing and protocol metadata collection from publicly accessible services. We do not attempt authentication or vulnerability exploitation. Because client IP addresses are considered Personally Identifiable Information (PII), all IPs are anonymized using the CryptoPan algorithm [13]. Our analysis focuses solely on publicly exposed services and does not track individual users.

For passive DNS measurements, IP addresses are likewise treated as PII, and queried domains may reveal Sensitive Personal Information (SPI). To mitigate these risks, we anonymize all client IPs using the same CryptoPan algorithm and limit domain analysis to aggregated statistics at the SLD granularity.

Overall, our measurements comply with ethical research guidelines and pose no significant risk to scanned infrastructure, network operators, or end users.

B Geographic Distribution of DNS Queries

Figure 8 shows the geographic distribution of DNS queries observed during the first three months (November 2023–January 2024) and the final three months (January–March 2025) of our measurement period. The comparison highlights both regional shifts in user activity and the rapid global expansion of the Starlink network. Over time, we observe increased adoption in historically underserved regions, particularly across parts of South America and Africa, reflecting Starlink's growing footprint in areas with limited terrestrial connectivity.

C DNS Leakage Statistics per Starlink PoP

We select a set of representative PoPs and summarize their observed DNS leakage in Table 6. The table reports the number of leaked private and public IP addresses, along with the corresponding private subnet prefixes, for each PoP.

The volume of leaked records partially reflects the extent of internal information exposure at a PoP. For example, jtnaidn2 leaks 70 private IP addresses, suggesting relatively weak information isolation, which is consistent with the asset scanning results discussed in Section 6.1. At the same time, leakage volume may also correlate with the scale and functional complexity of a PoP. As one of Starlink's major European PoPs, frntdeu1 leaks 66 private IP addresses, likely reflecting its large network footprint and rich internal service composition.

D Inference for Selected Naming Patterns

In this section, we detail the inference process for several representative DNS naming patterns observed in Starlink's infrastructure.

- (1) **SPI** — Service Plane Internal vs. Stateful Packet Inspection. The interpretation Service Plane Internal aligns more closely with terminology commonly used in large-scale carrier networks. All domains containing spi resolve exclusively to private IP addresses, and we do not observe combinations such as spi-fw1 (firewall). These observations support the conclusion that spi denotes internal service-plane components rather than packet inspection functionality.
- (2) **SPX** — Service Plane External vs. Service Plane Exchange. In contrast to spi, all spx-labeled domains resolve to public IP addresses and are frequently associated with fw1 (firewall) components. Given the complementary semantics of spi and spx, we infer that spx denotes externally facing service-plane elements exposed through firewall boundaries.
- (3) **PDU** — Power Distribution Unit vs. Protocol Data Unit. Although Protocol Data Unit is a standard term in networking theory, the observed deployment pattern strongly favors

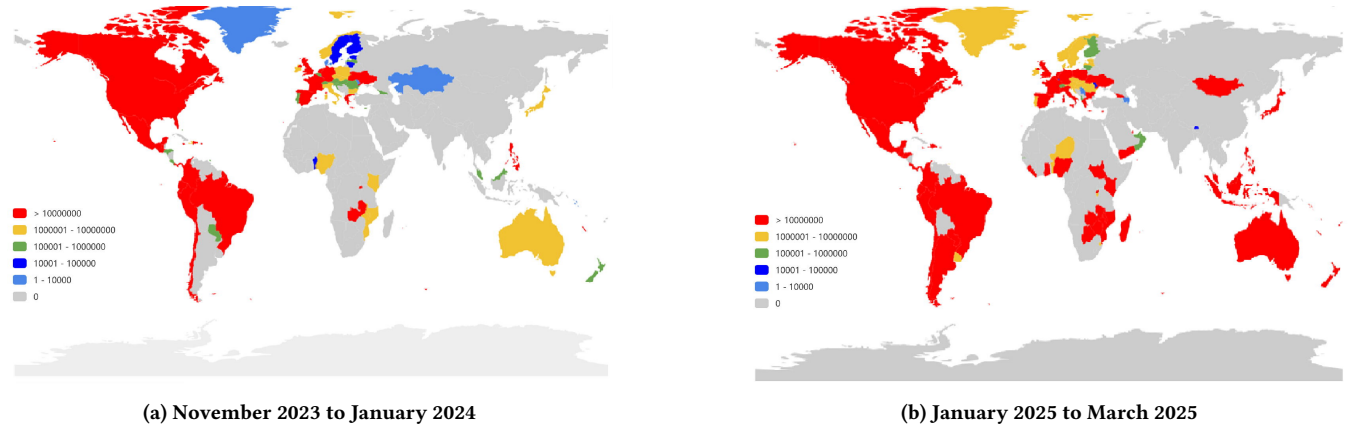


Figure 8: Geographic distribution of DNS queries during the first and last three months of the measurement period.

Table 6: Overview of DNS leakage across selected Starlink PoPs.

PoP	Private IP Count	Public IP Count	Private IP Prefixes
dnvrcox1	62	8	172.17.16.0/24,172.19.3.0/24,172.19.4.0/23
dohaqt1	22	3	172.17.46.0/24,172.19.1.0/24,172.19.2.0/24
frntdeu1	66	10	172.17.12.0/24,172.19.3.0/24,172.19.4.0/23
jhgazaf1	46	2	172.17.55.0/24,172.19.3.0/24,172.19.4.0/24
jtnaidn1	16	6	172.17.11.0/24,172.19.1.0/24,172.19.2.0/24
jtnaidn2	70	7	172.17.38.0/24,172.19.3.0/24,172.19.4.0/23
kentwax1	22	2	172.17.37.0/24,172.19.1.0/24,172.19.2.0/24
limaper1	56	6	172.17.17.0/24,172.19.3.0/24,172.19.4.0/24
lndngbr1	57	8	172.17.13.0/24,172.19.3.0/24,172.19.4.0/24
lsancax1	64	8	172.17.2.0/24,172.19.2.0/23,172.19.4.0/23
lsancax2	2	3	172.17.251.0/24
mdrdesp1	56	8	172.17.18.0/24,172.19.3.0/24,172.19.4.0/24
mlbeaus1	46	2	172.17.57.0/24,172.19.3.0/24,172.19.4.0/24
mlnnta1	46	2	172.17.53.0/24,172.19.3.0/24,172.19.4.0/24

Power Distribution Unit. Specifically, pdu devices consistently appear in redundant pairs (e.g., pdu1a and pdu1b), which is characteristic of physical power infrastructure in data centers.

- (4) **PKP** — Packet Processing vs. Peering Platform. The pkp label consistently co-occurs with srv, indicating a server-based role. Its widespread presence suggests functionality related to large-scale packet handling and traffic processing, which aligns with Starlink’s operational requirements. In contrast, a Peering Platform would typically correspond to exchange points rather than numerous individually numbered servers.

E IoT Device Types in Starlink

This section presents the types and relative proportions of IoT devices observed within the Starlink network, as shown in Figure 9. Video surveillance devices constitute the largest category, indicating that improvements in Starlink’s service quality increasingly support bandwidth-intensive applications. In addition, a non-negligible fraction of exposed IoT devices corresponds to industrial control systems (ICS), which may introduce elevated security risks due to their operational sensitivity and criticality.

F DNS Performance

We evaluate DNS resolution performance in the Starlink environment using globally distributed RIPE Atlas probes [51] operating within AS14593 and AS45700. Each probe issues repeated DNS queries for ten popular domains from the Tranco list [34] to (i) its default resolver and (ii) seven public resolvers. This setup allows us to approximate real-world resolver configurations among Starlink users.

Figure 10 compares resolution latency across regions. *Cloudflare DNS* consistently achieves the lowest average RTT and variance in all tested regions, indicating superior performance and stability. Although *Google DNS* is widely used, it shows higher latency variability, primarily due to the 8.8.4.4 instance. China-based resolvers such as *114DNS* and *BaiduDNS* perform poorly, likely because of geographically limited deployments. *Default* resolvers exhibit moderate latency but weaker stability overall. Among all resolvers, *Cloudflare DNS* performs best.

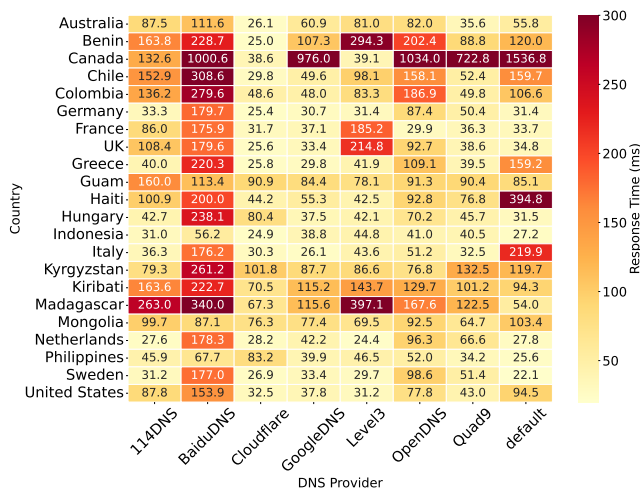


Figure 10: Regional performance of DNS resolvers.

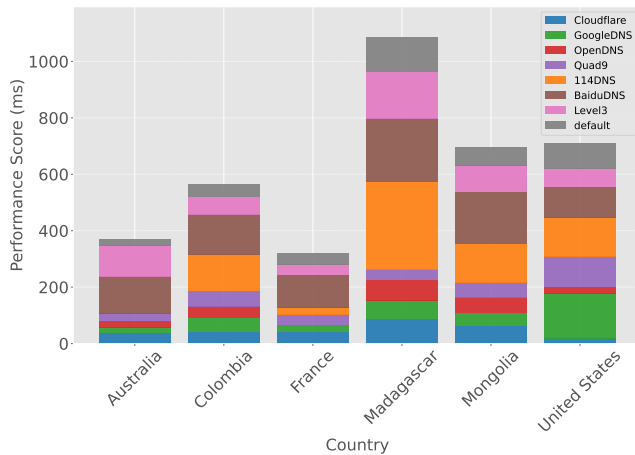


Figure 11: Resolution quality comparison across different DNS resolvers.

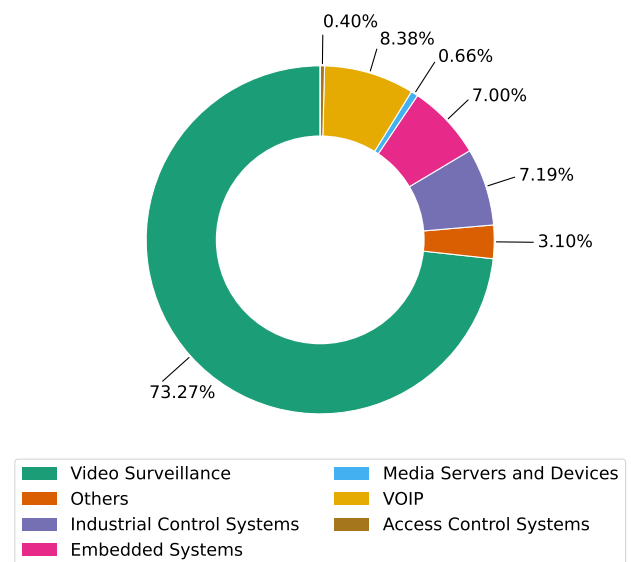


Figure 9: Distribution of IoT device types within the Starlink network.

Figure 11 evaluates DNS-to-CDN mapping performance by measuring the end-to-end latency of the returned CDN servers. *OpenDNS* and *Cloudflare DNS* consistently achieve the lowest latency. Major public resolvers, *OpenDNS*, *Cloudflare*, *Google DNS*, and *Quad9*, all outperform *default* resolvers, suggesting suboptimal CDN selection for users relying on default settings. In contrast, *114DNS* and *BaiduDNS* frequently return non-optimal CDN endpoints, likely due to resolver-centric routing decisions rather than client-centric ones.

For example, in *Madagascar*, the *default* resolver often maps *youtube.com* to U.S.-based servers, whereas *Quad9* resolves it to servers in *South Africa*, yielding significantly lower latency. Prior studies [48] have shown that such DNS-CDN geographical mismatches are prevalent in satellite communication environments, which disrupts the proximity-based content delivery strategy originally designed by CDNs, thereby exacerbating the performance bottlenecks of high-latency satellite links.